

AKINTAYO Sofiyah Modupeola

Cybersecurity Engineer & Data Analyst/Scientist

📍 Lagos, NG sofiyah-akintayo-7096b9122

ABOUT

I am an accomplished Medical Scientist and GRC Engineer with 4 years of experience in data analytics, transitioning into Cyber space, AI and Data Science. Proficient in extracting and interpreting complex datasets while ensuring rigorous technical governance, risk management, and regulatory compliance. Adept at applying generative AI principles for innovative solutions, backed by a deep understanding of secure data frameworks. Eager to leverage analytical rigor and GRC expertise to develop ethical, secure, and impactful AI solutions.

WORK EXPERIENCE

Blue Engineering Trainee | International Cybersecurity and Digital Forensic Academy | Student Trainee

Jul 2025 - Present, Nigeria

- **Compliance Documentation & Reporting System Design:** Architected a centralized, data-driven compliance system for TechFlow Industries, transitioning the firm from reactive paper-based processes to a proactive digital repository with standardized reporting templates and a real-time KPI dashboard.
- **Authorized External Security & Compliance Assessment:** Conducted a comprehensive vulnerability assessment using Nmap and Burp Suite Professional, identifying critical flaws like SQL Injection and Broken Access Control, and mapped findings to the OWASP Top 10 framework for executive remediation.
- **Quantitative Risk Analysis & Treatment Planning :** Conducted financial risk modeling for mobile banking vulnerabilities, calculating Single Loss Expectancy (SLE) and Annualized Loss Expectancy (ALE) to demonstrate a 90% risk reduction and a high ROI for proposed security controls.
- **HIPAA GRC Engagement:** Performed a gap analysis against the HIPAA Security Rule, designed a formal Password and Authentication Policy (PAP), and developed a ransomware recovery strategy involving forensic analysis and sanitized system restoration.
- **Cybersecurity Risk Assessment & Management Strategy:** Developed a hybrid risk management framework for TechStart Solutions by integrating NIST and ISO 31000 standards, producing a comprehensive risk register and a three-phase remediation roadmap for 500+ clients.
- **Multinational Compliance Crisis Management (GlobalSync Inc.):** Assumed the role of Chief Compliance Officer to resolve complex international issues including GDPR data transfer violations, FCPA bribery allegations, and the implementation of a "Tone at the Top" governance strategy.
- **Security Log Analysis & Threat Detection:** Analyzed failed authentication and vulnerability logs to detect brute-force attacks and unaddressed critical vulnerabilities (e.g., CVE-2023-9012), proposing technical remediations like MFA and automated SIEM flagging.
- **Linux Security Monitoring & Auditing:** Executed hands-on system auditing and hardening on Linux environments using tools such as auditd, Lynis, and journalctl to identify and remediate configuration vulnerabilities.
- **Ransomware Incident Response Planning (FinanceFirst Credit Union):** Developed a detailed ransomware incident response playbook, establishing critical protocols for initial detection, system isolation, log preservation, and secure credential management to ensure business continuity.
- **Digital Transformation Risk Assessment (SecureBrew Cafe):** Conducted a comprehensive risk analysis for a cloud-based POS and loyalty app ecosystem; identified high-impact risks like Man-in-the-Middle attacks and recommended isolation of public Wi-Fi from secure transaction networks.

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- Continuous Risk Monitoring & Detective Control Implementation: Designed and deployed a detective control using the Wazuh platform to monitor a designated sensitive file directory for unauthorized modifications, ensuring a centralized audit trail and real-time alerting for compliance mandates.
- Disaster Recovery After-Action Analysis (FinanceFirst Bank): Authored a formal after-action report following a primary data center failure; analyzed the execution of disaster recovery protocols, data synchronization integrity, and stakeholder communication effectiveness to improve future resilience.
- Crisis Communication & Data Breach Response Simulation: Orchestrated a simulated data breach response including the development of a Crisis Communication Strategy, employee notification plans, and media response statements to maintain organizational reputation and transparency.
- Developed a specialized Python-based tool to automate Governance, Risk, and Compliance (GRC) workflows, transitioning qualitative risk assessments into data-driven financial models.

Technical Trainee - Data Science and Machine Learning | NITDA / Federal Government of Nigeria

Feb 2025 - Present, Lagos, Lagos State, Nigeria

Undergoing hands-on fellowship focused on technical specialization, project delivery, and workplace readiness within Data Science and Machine Learning.

- Applied SQL, Python, and Azure Machine Learning services to develop real-world solutions within the African Deeptech Challenge, demonstrating practical application of advanced AI/ML concepts.
- Undergoing hands-on technical specialization and project delivery, focusing on data science and machine learning, preparing for immediate workplace readiness.
- Engaged in a rigorous fellowship program, building foundational expertise in AI/ML through practical assignments and collaborative problem-solving.
- Applied Python in Breast Cancer Classification using different classification models.
- Engineered a predictive financial tool using Python to model cybersecurity risk impact, utilizing various data points such as Single Loss Expectancy (SLE) and Annual Rate of Occurrence (ARO).
- Designed a user-centric interface that allows for dynamic "What-If" scenario modeling, helping organizations visualize the financial benefit of specific security controls before deployment.

Cybersecurity Intern | Secure Edge Technologies - LASGIPP

Nov 2024 - Jan 2025, Lagos, Lagos State, Nigeria

Developed audit plans and internal auditing processes, authored cybersecurity newsletters, and significantly enhanced organizational security awareness.

- Developed comprehensive audit plans and internal auditing processes, strengthening organizational security posture and compliance.
- Authored impactful cybersecurity newsletters, significantly increasing employee awareness and adherence to best practices across the organization.
- Contributed to enhanced overall organizational security and awareness through proactive development and dissemination of critical cybersecurity information.

Technologist | Eko University of Medicine and Health Sciences

Dec 2020 - May 2024, Lagos, Lagos State, Nigeria

Managed the Morbid Anatomy Museum and the University's Histopathology Laboratory, ensuring efficient operations and informing improvements.

- Managed daily operations of the Morbid Anatomy Museum and Histopathology Laboratory, ensuring efficient resource utilization and compliance with academic standards.

- Analyzed end-of-posting evaluation sheets to identify operational inefficiencies and inform strategic improvements in departmental effectiveness.
- Streamlined laboratory workflows and museum processes, contributing to optimized operational efficiency and improved data management.

Technical Trainee - Animation | NITDA / Federal Government of Nigeria

Dec 2022 - Apr 2023, Lagos, Lagos State, Nigeria

Completed hands-on training in animation, creating explainer and animated videos using various tools.

- Utilized various animation tools to create diverse explainer and animated videos, enhancing communication and visual storytelling capabilities.
- Completed intensive hands-on training in animation techniques, mastering software and workflows for multimedia content creation.

EDUCATION

Cell Biology and Genetics | 4.29/5.00 | University of Lagos | M. Sc.

Lagos, Lagos State, Nigeria, Sep 2018 - Sep 2019

Anatomy | 4.35/5.00 | Ladoke Akintola University of Technology | B. Tech.

Ogbomosho, Oyo State, Nigeria, Sep 2010 - Sep 2015

SKILLS

Data Analytics | Data Analytics, Data Interpretation, Data Visualization, Complex Datasets

Artificial Intelligence | Generative AI, AI Fundamentals, AI Solutions, Azure AI Services

Machine Learning | Machine Learning, Azure Machine Learning

Programming Languages | Python, SQL

Data Tools & Platforms | Power BI, Microsoft Excel, Microsoft Azure AI services, Linux

Animation & Design | Animation, Explainer Videos, Adobe AfterEffect, Adobe Illustrator, Alice

Microsoft Office Suite | MS Word, Excel, PowerPoint

Soft Skills | Attention to Detail, Problem-Solving, Critical Thinking, Communication & Storytelling, Adaptability & Flexibility, Curiosity, Continuous Learning, Collaboration, Teamwork

Vendor & Third-Party Risk Management (TPRM)

Cybersecurity Technical Tools and Platforms | Security Monitoring & SIEM: Wazuh, Vulnerability Scanners: Burp Suite Professional, Nmap, Nikto, and Nuclei., Linux Auditing & Log Analysis, Netdiscover, Cloud & Virtualization: VirtualBox, VMware, and Ubuntu Server environments.

Risk Assessment & Modeling | Quantitative Risk Analysis, Financial Impact Analysis, Qualitative Assessment

Governance & Policy Design | Policy Development, Compliance Framework Mapping, Framework Implementation

Incident Management & Resilience | Incident Response (IR), Business Continuity & Disaster Recovery (BCDR), Crisis Communication

CERTIFICATES

ISO 27001 & ISO 42001 Lead Implementer | International Cybersecurity and Digital Forensic Academy

Dec 2025

Microsoft Certified: Azure AI Fundamentals | Microsoft

Jul 2025

Data Science/Machine Learning | DeepTech Ready by Data Science Nigeria

Dec 2025

ISO/IEC 20000:2018 I. T Service Management Systems (ITMS) Foundation | SandBP

Jan 2024

Career Essentials in Generative AI | Microsoft and LinkedIn
Jan 2024

ISO/IEC 22301:2019 Business Continuity Management
Systems (BCMS) Foundation | SandBP
Jan 2024

Google Data Analytics | Google
Jan 2023

LANGUAGES

English